

Classical Mechanics

Lecture 7: Liouville's Theorem and Hamiltonian Flow

Lectures by Leonard Susskind

Transcript-derived companion notes curated by [LazyingArt LLC](#)

Produced with [Video2Book](#)

Chapter 1

Liouville's Theorem and Hamiltonian Flow

Overview. Hamilton's equations do more than repackage the equations of motion. They define a velocity field on phase space whose divergence vanishes. That simple cancellation is Liouville's theorem: Hamiltonian evolution preserves phase-space volume. We shall reach it in the order in which its meaning becomes visible—first energy conservation, then constant-energy surfaces, then a fluid of possible initial conditions, and finally the contrast between volume-preserving flow and a dissipative sink. The same language leads naturally to Poisson brackets.

1.1 Why Another Form of Mechanics?

The theorem we want is the continuous classical counterpart of the reversible state diagrams introduced at the beginning of the course. In a reversible discrete law, every state has one arrow entering and one arrow leaving. The present is enough to determine both the next state and the preceding one. A bad law sends many states into one sink: the future may be determined, but the past is no longer recoverable.

Liouville's theorem says what becomes of that distinction when the set of states is a continuous phase space. The phrase replacing one arrow in and one arrow out will be *as much flow in as flow out*. Before that phrase can be made mathematical, however, we need to reconstruct the flow from Hamilton's equations.

There is a historical puzzle in doing so. Hamilton, Jacobi, Lagrange, and Poisson built elaborate and beautiful forms of classical mechanics without knowing the quantum theory in which those forms would become indispensable. Newton had a direct target: planetary motion. Lagrange and Hamilton also had physical problems before them, but they pursued analogies and structures far beyond any one orbit. Stationary action in mechanics resembled least-time principles in optics; ray motion and particle motion could be put into parallel mathematical forms. They simplified, rearranged, and unified the equations because the structures themselves appeared to contain physics.

We now know that this instinct was extraordinarily productive. Logically, many features of classical mechanics can be understood as limits of quantum mechanics. Historically, the order was reversed: the classical structures were found first, and only much later did their deeper use become clear. A provocative joke attributed to Feynman made the point by contrasting mathematics pursued

internally with physics constrained by an external partner, the physical world. The joke is not an argument, but the distinction is useful: formal beauty matters here because nature repeatedly selects these formal structures.

Question from the audience. Did Newton have the modern concept of kinetic energy, and where did the concept of energy begin? ^a

Response. Newton's mechanics was organized around force and motion, not around a mature general energy principle. The exact historical priority is not settled in the discussion. Leibniz's *vis viva*, $\sum_i m_i v_i^2$, was an important predecessor and equals twice the modern kinetic energy. Later work on heat made the conversion between heat and mechanical work central. Early caloric theories treated heat as a conserved substance, so the modern idea of interchangeable forms of energy emerged only gradually.

^aLecture timestamp: 00:04:24–00:05:50.

1.2 Hamilton's Equations and Energy Conservation

Let a system have n canonical pairs,

$$(q_i, p_i), \quad i = 1, \dots, n. \quad (1.1)$$

There is one momentum for each coordinate and one coordinate for each momentum. The law of motion is packaged in one function,

$$H = H(q_1, \dots, q_n, p_1, \dots, p_n, t), \quad (1.2)$$

and Hamilton's equations are

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.} \quad (1.3)$$

The sign is part of the canonical structure. The q -equation carries the plus sign; the p -equation carries the minus sign.

Although the equations come in indexed pairs, the dynamics need not separate pair by pair. The Hamiltonian may couple every q_i and p_i , so $\dot{p}_1$, for example, may depend on the state of the entire system.

Question from the audience. Is the presentation now axiomatic? What relation between p and q has been assumed? ^a

Response. Hamilton's equations define the formal theory once $H(q, p)$ and the canonical pairs are specified. There is no universal extra assumption $p = m\dot{q}$. That relation holds for an ordinary Cartesian particle with quadratic kinetic energy, but other coordinates and other Hamiltonians give other momentum–velocity relations. The equations themselves supply the relation through $\dot{q}_i = \partial H / \partial p_i$. Whether a proposed Hamiltonian is realized in nature is a further physical question.

^aLecture timestamp: 00:17:08–00:18:18.

We can now recover energy conservation in one line of reasoning. Along a trajectory, the total derivative of H is

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t}. \quad (1.4)$$

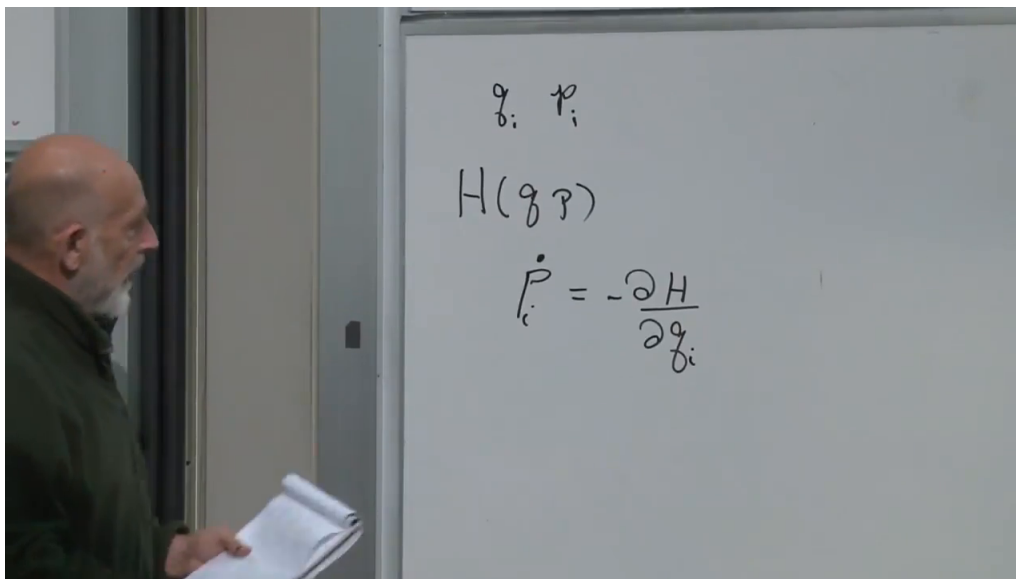


Figure 1.1: Canonical variables, the function $H(q, p)$, and the momentum-side Hamilton equation at the beginning of the recap.

Lecture frame: 00:07:29.

The last term means explicit time dependence at fixed q and p , perhaps because an externally prescribed parameter is changing. Substituting Eq. (1.3),

$$\frac{dH}{dt} = \sum_i \left[\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right] + \frac{\partial H}{\partial t} \quad (1.5)$$

$$= \frac{\partial H}{\partial t}. \quad (1.6)$$

Every term in the sum cancels its partner. Consequently,

$$\frac{\partial H}{\partial t} = 0 \quad \implies \quad \frac{dH}{dt} = 0. \quad (1.7)$$

For a closed, time-translation-invariant system, the Hamiltonian is conserved and is the energy.

Question from the audience. What changes if the Hamiltonian depends explicitly on time?
^a

Response. Then Eq. (1.6) still holds, but the subsystem energy need not be constant. An explicit time dependence usually represents an omitted driver exchanging energy with the system. If that driver is promoted to a dynamical part of a larger closed system, the larger Hamiltonian can again be time independent and its total energy conserved.

^aLecture timestamp: 00:10:32–00:12:33.

1.3 Energy as Geometry in Phase Space

Suppose now that H has no explicit time dependence. The initial condition fixes a number E , and the phase point remains on

$$H(q, p) = E. \quad (1.8)$$

Phase space has $2n$ dimensions. At regular points, one equation among its coordinates defines a $(2n - 1)$ -dimensional hypersurface. Different values of E give different hypersurfaces. They need not be closed or compact; energy conservation says only that a trajectory beginning on one of them never leaves it.

For the adapted harmonic oscillator from the preceding lecture,

$$H = \frac{\omega}{2}(p^2 + q^2). \quad (1.9)$$

Thus each constant-energy set is a circle,

$$p^2 + q^2 = \frac{2E}{\omega}. \quad (1.10)$$

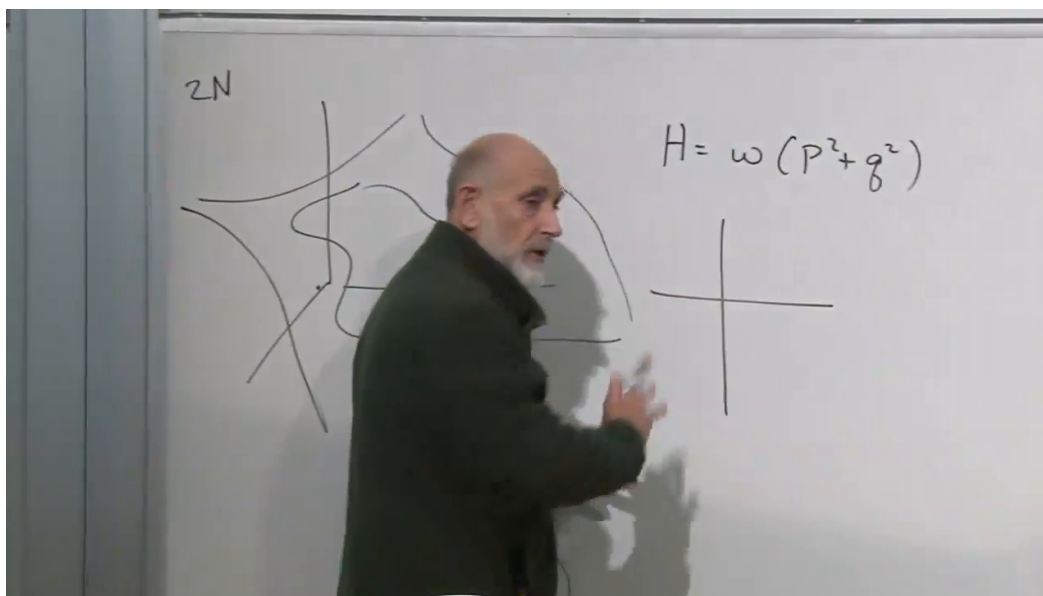


Figure 1.2: The $2n$ -dimensional phase-space sketch and the harmonic oscillator Hamiltonian used to introduce constant-energy surfaces.

Lecture frame: 00:15:20.

The equations of motion,

$$\dot{q} = \omega p, \quad \dot{p} = -\omega q, \quad (1.11)$$

make the point move clockwise around its circle. The energy equation tells us which curve contains the motion; Hamilton's equations tell us the direction and speed along that curve.

Question from the audience. Are Hamilton's two equations and $H = E$ several versions of the same equation? ^a

Response. No. Hamilton's equations are the two first-order laws of motion for one canonical pair. They imply $dH/dt = 0$, from which $H = E$ follows. The value of E is selected by the initial condition. The equation $H = E$ identifies a surface; it does not by itself specify the velocity along that surface.

^aLecture timestamp: 00:15:53–00:17:06.

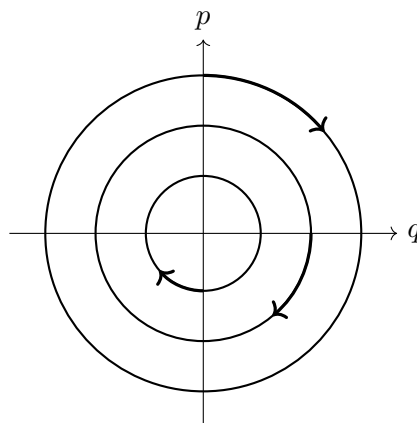


Figure 1.3: Constant energy gives the circles; the oscillator equations give the clockwise tangent flow.

1.4 From One State to a Flow of States

Instead of following one initial condition, sprinkle phase space with every possible initial condition and start the clock. Each point is a complete possible state, and each state moves according to Hamilton's equations. The sprinkling can be imagined as a dust or an imaginary fluid. Nothing material is being poured into phase space; the fluid is a way to see all trajectories at once.

The phase-space velocity field is

$$\mathbf{V}(q, p, t) = (\dot{q}_1, \dots, \dot{q}_n, \dot{p}_1, \dots, \dot{p}_n). \quad (1.12)$$

For one canonical pair we write

$$V_q = \dot{q} = \frac{\partial H}{\partial p}, \quad V_p = \dot{p} = -\frac{\partial H}{\partial q}. \quad (1.13)$$

At each point the Hamiltonian assigns a vector telling the phase point where to go next. If H is time independent, the vector field is stationary: returning to the same phase point gives the same velocity. If H depends on time, the vector field itself changes with time, but it is still defined at each instant.

For the harmonic oscillator every radius turns with angular speed ω . Points near the origin have smaller linear speed because they travel a shorter circle in the same period. The entire dust therefore rotates as a rigid carousel.

Question from the audience. Does rigid rotation mean that all oscillator trajectories have the same period? ^a

Response. Yes. That amplitude independence is the defining special property of the harmonic oscillator. Every circle is traversed with the same angular frequency ω , even though the magnitude of the phase-space velocity is smaller on a smaller circle.

^aLecture timestamp: 00:21:10–00:21:42.

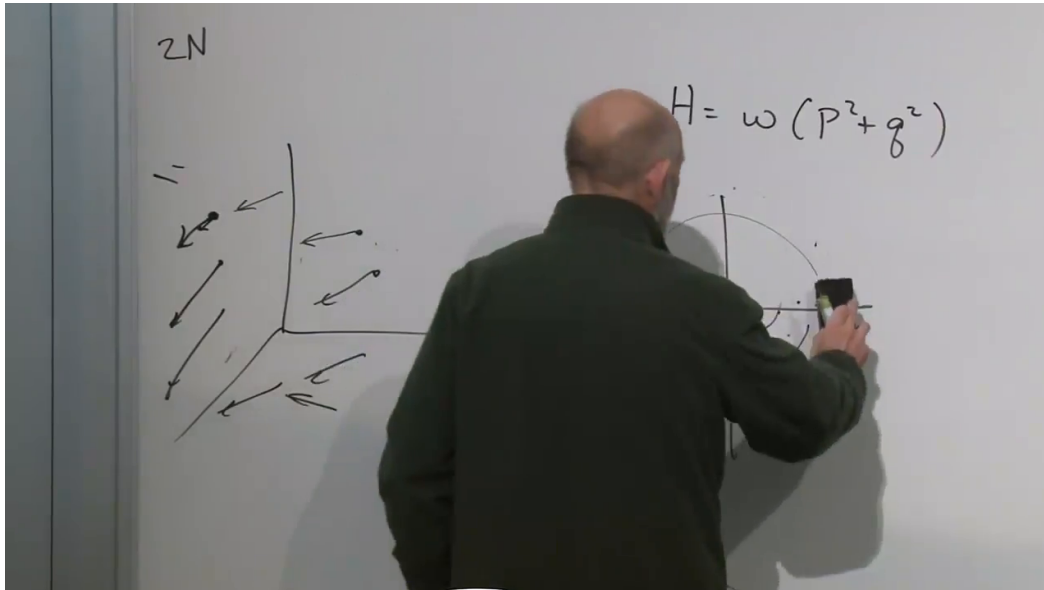


Figure 1.4: The harmonic-oscillator velocity field. The vectors are tangent to the circles and become shorter toward the origin.

Lecture frame: 00:24:50.

Question from the audience. Is the phase-space velocity the same at every point? ^a

Response. No. Both its direction and magnitude generally vary with position. For the oscillator the vector is tangent to the local circle and has magnitude $\omega\sqrt{p^2 + q^2}$. A vector field means one assigned vector at every point, not one identical vector copied everywhere.

^aLecture timestamp: 00:24:29–00:25:37.

Question from the audience. What exactly are the particles and density of this imaginary fluid? ^a

Response. Each point is one complete state of the whole mechanical system. It is not one physical particle. Five particles moving in one spatial dimension require five positions and five momenta, hence a ten-dimensional phase space; in three dimensions they require thirty dimensions. The dust is an ensemble of alternative states. If the fluid picture is distracting, simply take a region of initial conditions, evolve every point in it, and compare its phase-space volume before and after.

^aLecture timestamp: 00:56:03–00:58:40.

1.5 Compression and Divergence

To ask whether this imaginary fluid loses volume, begin in one dimension. Let points on a line move with velocity $v(x)$. Two nearby points separated by δx have a velocity difference

$$\delta v \simeq \frac{dv}{dx} \delta x. \quad (1.14)$$

If $dv/dx > 0$, the point ahead moves faster and the interval stretches. If $dv/dx < 0$, it compresses. A stationary one-dimensional incompressible flow therefore obeys

$$\frac{dv}{dx} = 0. \quad (1.15)$$

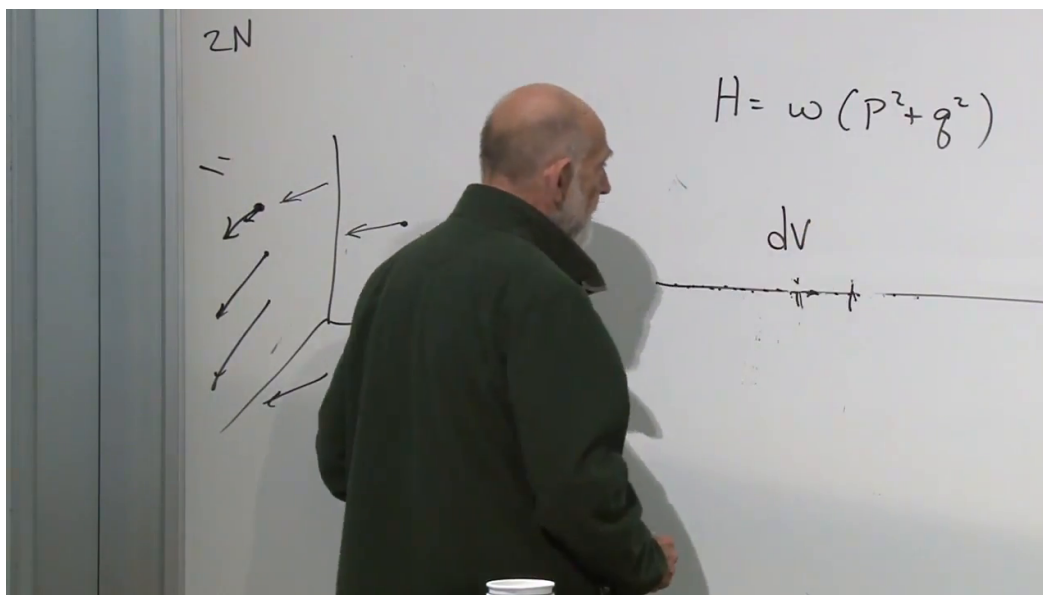


Figure 1.5: The one-dimensional construction leading to dv/dx as the local stretching or compression rate.

Lecture frame: 00:30:20.

There are two equivalent viewpoints. We may follow a material interval and ask whether its length changes. Or we may hold a small interval fixed and ask whether the amount entering per unit time equals the amount leaving. The first follows the fluid; the second watches it pass.

In more than one dimension a region may stretch along one axis and compress along another without changing total volume. For a velocity field $\mathbf{V} = (V_x, V_y, V_z)$, the relevant local rate is the divergence,

$$\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}. \quad (1.16)$$

Positive divergence is local expansion, negative divergence is local contraction, and incompressibility is

$$\nabla \cdot \mathbf{V} = 0. \quad (1.17)$$

Equal inflow and outflow through a fixed small box is the same local statement as preservation of the volume of a moving small box.

Question from the audience. Are we discussing a fluid in ordinary space or a flow in phase space? ^a

Response. The divergence argument is first a general fluid argument. We shall apply it to the even-dimensional phase space, where the coordinate directions are the q_i and p_i , and the velocity components are $\dot{q}_i$ and $\dot{p}_i$.

^aLecture timestamp: 00:34:27–00:34:40.

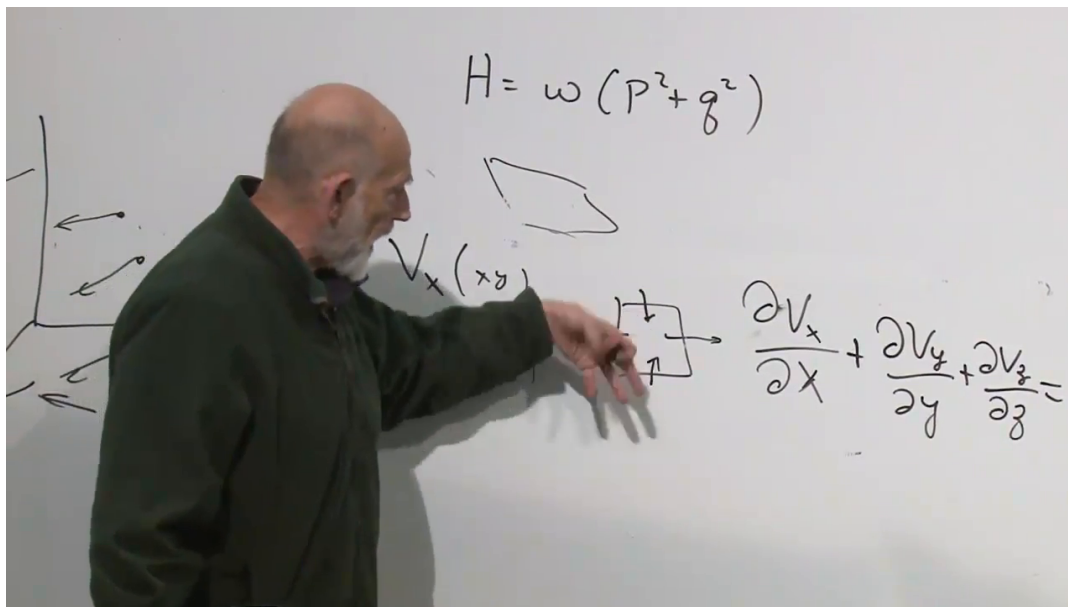


Figure 1.6: A two-dimensional box, its flow components, and the sum of same-coordinate derivatives that defines divergence.

Lecture frame: 00:35:30.

Now return to the discrete state maps. In a reversible cycle, every node has one arrow in and one arrow out, and a chosen collection of states retains the same number of members as it evolves. In a sink, many arrows enter and too few leave; several alternatives become indistinguishable.

The continuous question is now exact: does Hamiltonian flow have zero divergence, so that a small region of distinguishable states never gets squashed into a smaller phase-space volume?

1.6 Liouville's Theorem

Theorem 1.1 (Liouville's theorem). *For a smooth Hamiltonian system, the canonical phase-space velocity field has zero divergence:*

$$\sum_i \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) = 0. \quad (1.18)$$

Equivalently, Hamiltonian evolution preserves the canonical phase-space volume $dq_1 \cdots dq_n dp_1 \cdots dp_n$.

Question from the audience. Are the components called V simply the p -dots and q -dots? ^a

Response. Yes. For one pair, $V_q = \dot{q}$ and $V_p = \dot{p}$. Lining up all such derivatives gives the velocity vector of the flow in the full phase space.

^aLecture timestamp: 00:42:20–00:43:22.

For one canonical pair, Hamilton's equations give

$$V_p = -\frac{\partial H}{\partial q}, \quad V_q = \frac{\partial H}{\partial p}. \quad (1.19)$$

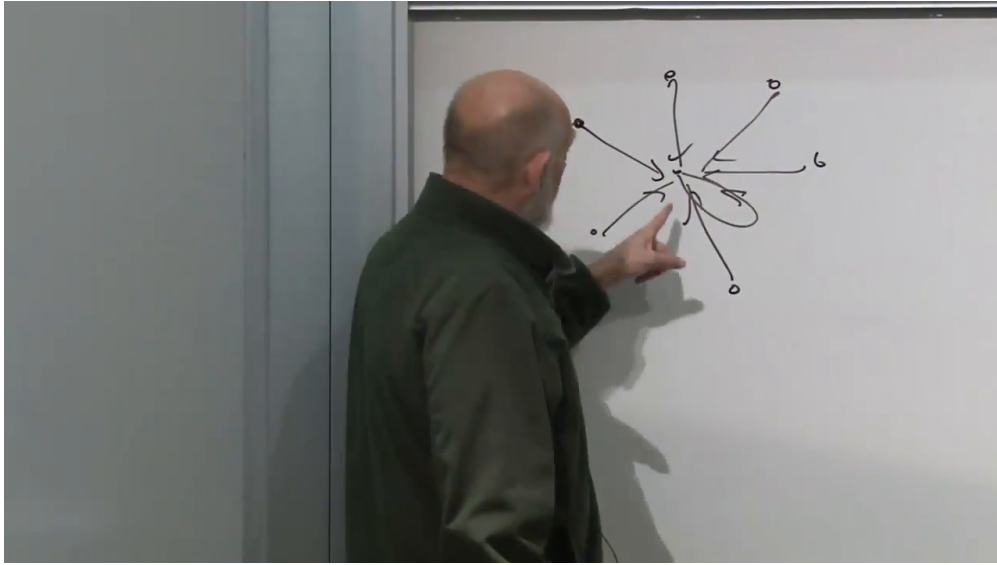


Figure 1.7: The discrete sink used to reconnect divergence with the first-lecture test for reversible laws.

Lecture frame: 00:40:15.

The divergence is therefore

$$\frac{\partial V_p}{\partial p} + \frac{\partial V_q}{\partial q} = -\frac{\partial}{\partial p} \left(\frac{\partial H}{\partial q} \right) + \frac{\partial}{\partial q} \left(\frac{\partial H}{\partial p} \right) \quad (1.20)$$

$$= -\frac{\partial^2 H}{\partial p \partial q} + \frac{\partial^2 H}{\partial q \partial p} = 0. \quad (1.21)$$

The equality of mixed partial derivatives can be seen with four nearby points A, B, C, D at the corners of a small rectangle in the q - p plane. Taking a q -difference and then a p -difference produces, apart from the two small side lengths,

$$F(B) + F(D) - F(A) - F(C). \quad (1.22)$$

Taking the differences in the opposite order produces the same combination. In the small-rectangle limit this is $\partial_p \partial_q F = \partial_q \partial_p F$, provided the required derivatives are continuous.

For many degrees of freedom, the same cancellation occurs for every canonical pair:

$$\sum_i \left(\frac{\partial \dot{p}_i}{\partial p_i} + \frac{\partial \dot{q}_i}{\partial q_i} \right) = \sum_i \left(-\frac{\partial^2 H}{\partial p_i \partial q_i} + \frac{\partial^2 H}{\partial q_i \partial p_i} \right) \quad (1.23)$$

$$= 0. \quad (1.24)$$

Stretching in one canonical direction is balanced by compression in its conjugate direction. This pairwise cancellation is stronger structure than a coincidental cancellation among unrelated axes.

Question from the audience. Is the summation over all canonical pairs understood? ^a

Response. Yes. Equation (1.24) is summed over i , and the two mixed derivatives cancel separately within each q_i, p_i pair.

^aLecture timestamp: 00:52:00–00:52:36.

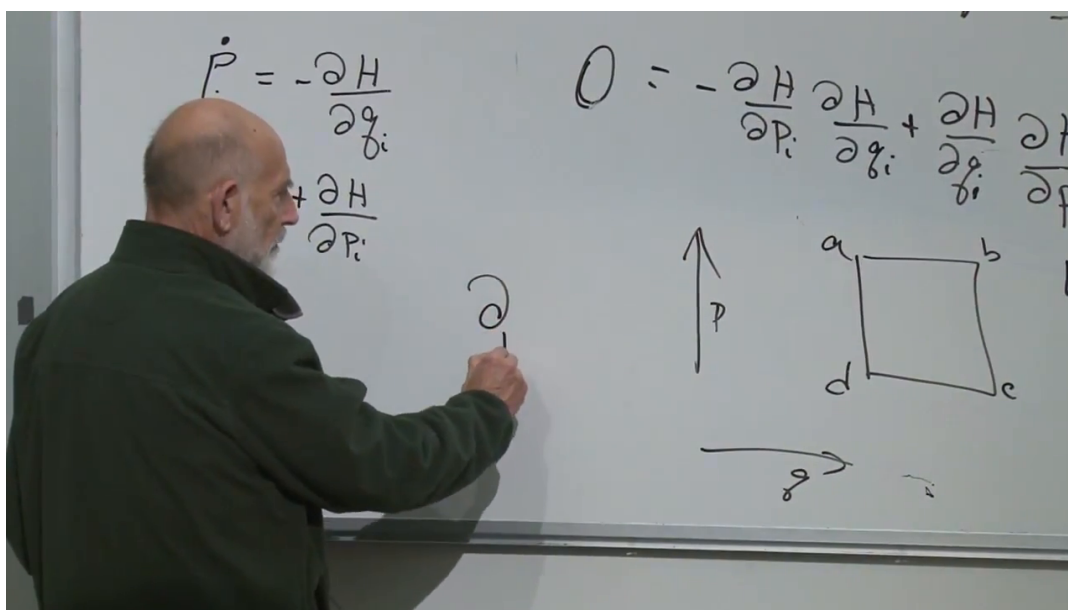


Figure 1.8: The finite-difference rectangle used to explain why smooth mixed partial derivatives commute.

Lecture frame: 00:47:30.

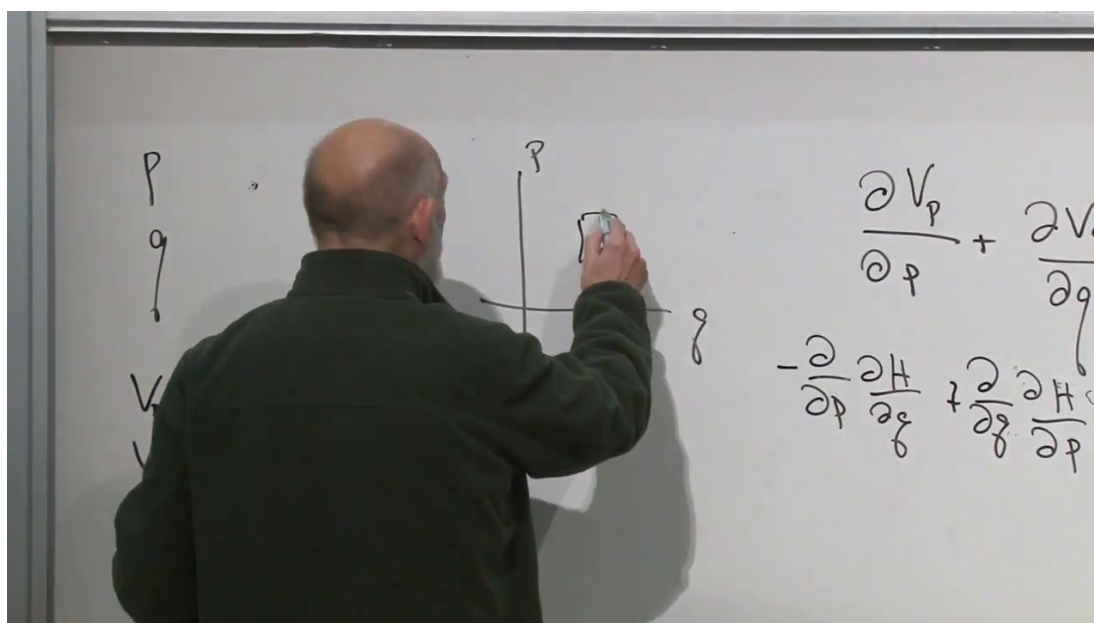


Figure 1.9: The canonical divergence after Hamilton's equations have reduced it to two mixed derivatives with opposite signs.

Lecture frame: 00:50:20.

Question from the audience. The displayed divergence looks Cartesian. Does the theorem survive a change of coordinates? ^a

Response. The natural statement is made in canonical coordinates and with the canonical volume element $dq_1 \cdots dq_n dp_1 \cdots dp_n$. A canonical, or symplectic, transformation changes coordinates and momenta together and preserves that volume. In arbitrary noncanonical coordinates, a Jacobian factor belongs in the volume measure; the naive product of coordinate increments need not be invariant.

^aLecture timestamp: 00:52:50–00:53:22, 00:58:18–00:58:40.

Question from the audience. Does Liouville's theorem require a time-independent Hamiltonian? ^a

Response. No. Energy conservation required $\partial H / \partial t = 0$, but the divergence proof differentiates only with respect to q_i and p_i at a fixed time. The mixed partials still cancel for $H(q, p, t)$. A time-dependent Hamiltonian therefore generates a time-dependent but instantaneously volume-preserving canonical flow.

^aLecture timestamp: 00:53:43–00:55:51.

1.7 What Volume Preservation Does and Does Not Mean

Take any small region R_0 of initial conditions and evolve every point for a time t . Liouville's theorem says

$$\text{Vol}(R_t) = \text{Vol}(R_0). \quad (1.25)$$

The region may move, rotate, shear, and stretch into a long filament. Its volume, not its shape, is protected. Distinct phase points cannot coalesce in a finite smooth Hamiltonian evolution.

For finite times, a smooth Hamiltonian flow is an invertible map with a smooth inverse obtained by evolving backward. This also explains the classroom claim that a connected blob cannot pinch into disconnected pieces: the flow deforms the region continuously and preserves its topology, even though an extremely thin neck may make two lobes look separated at finite resolution.

Question from the audience. Can a phase-space region pinch off and split into two pieces? ^a

Response. Not under a smooth deterministic Hamiltonian flow for a finite time. At a hypothetical pinch point there is still one phase-space point and one assigned velocity, not two incompatible directions. The region can develop a very long, narrow connecting filament, but the evolution remains one-to-one and does not tear it.

^aLecture timestamp: 01:04:10–01:05:49.

Question from the audience. Are two widely separated lobes of a stretched blob physically connected or quantum-entangled? ^a

Response. No such physical connection is implied. The blob is a set of alternative initial conditions, not one material object. Calling it one region records the common set from which those states evolved; it says nothing about forces or quantum entanglement between them.

^aLecture timestamp: 01:08:01–01:09:16.

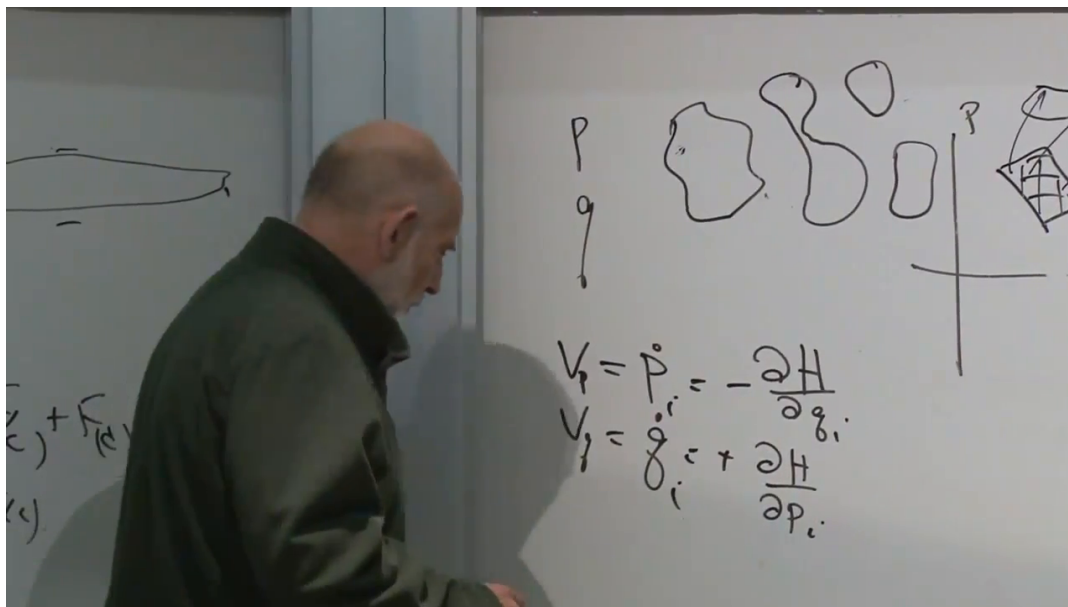


Figure 1.10: Possible deformations of one phase-space region. A smooth Hamiltonian flow may produce a thin neck, but not a genuine finite-time break into independently evolved pieces.

Lecture frame: 01:04:55.

Question from the audience. What happens near the upright unstable equilibrium of a rigid pendulum? Do clockwise and counterclockwise trajectories overlap? ^a

Response. Nearby initial conditions can separate onto very different trajectories, producing the thin filaments just described. They do not become the same phase point: the sign of the infinitesimal displacement or momentum distinguishes the two sides. The exactly balanced state has its own uniquely determined motion, namely remaining at equilibrium in the ideal model.

^aLecture timestamp: 01:07:07–01:07:47.

1.7.1 Expansion in Position Space and Contraction in Momentum Space

If a canonical transformation expands a range of positions by a factor a , the conjugate momentum scale contracts by the reciprocal factor:

$$q \longrightarrow aq, \quad p \longrightarrow \frac{p}{a}, \quad dq dp \longrightarrow dq dp. \quad (1.26)$$

This is the clean phase-space content behind the gas-expansion example. In an adiabatic expansion, the accessible ordinary-space region grows while typical momenta fall and the gas cools. A detailed thermodynamic relation depends on dimension, the expansion protocol, and interactions; Liouville's theorem supplies the volume constraint, not every thermodynamic coefficient.

Question from the audience. Does the same reasoning apply to an expanding universe, even for a sudden expansion? ^a

Response. Liouville's theorem applies whenever the complete dynamics is Hamiltonian, including explicitly time-dependent canonical variables. Expansion in position directions must be accompanied by the corresponding contraction in canonical-momentum directions. A cosmological scale factor, a controlled expansion of a gas, and a freely exploding cloud are not the same physical process, however. A Hubble-like correlation between distance and recession speed can arise because faster objects travel farther, whereas adiabatic cooling concerns contraction of the local momentum distribution. The theorem constrains the full phase-space map in either case.

^aLecture timestamp: 00:53:23–00:56:00.

Question from the audience. What practical use does Liouville's theorem have? ^a

Response. It is central to statistical mechanics. It controls how ensembles of microscopic states evolve and underlies the relation between phase-space volume and entropy. The expansion and cooling of a gas is one immediate illustration; later it becomes a basic tool for equilibrium and nonequilibrium reasoning.

^aLecture timestamp: 01:06:16–01:07:04.

1.8 A Flow That Changes Shape Violently

The harmonic oscillator preserves both area and shape because its flow is a rigid rotation. That is exceptional. To separate volume preservation from rigidity, take

$$H = \lambda pq, \tag{1.27}$$

where λ has units of inverse time. The lecture sets $\lambda = 1$; retaining it makes the dimensions explicit.

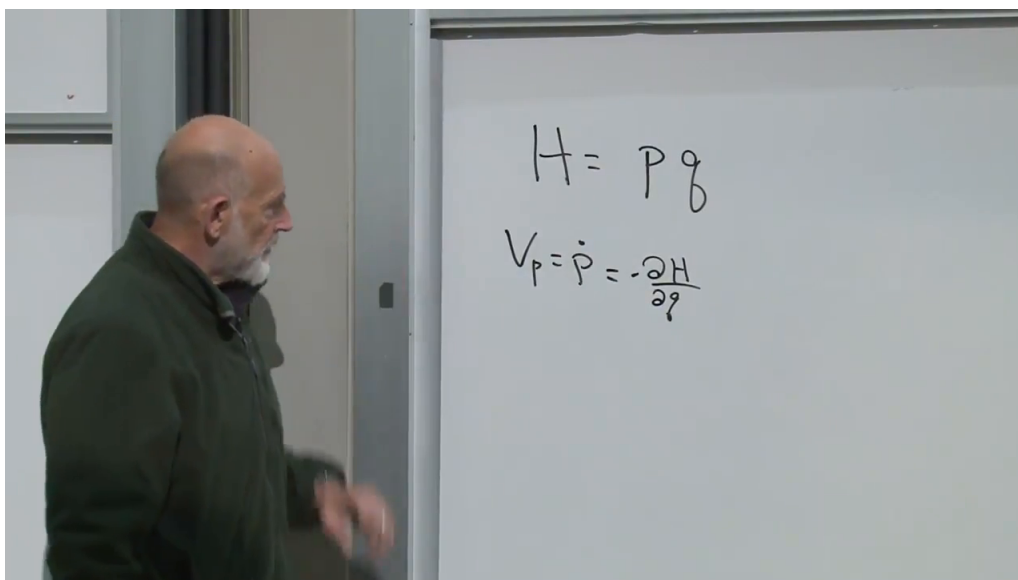


Figure 1.11: The beginning of the $H = pq$ example and the first phase-space velocity component.

Lecture frame: 01:00:11.

Hamilton's equations are

$$\dot{p} = -\lambda p, \quad \dot{q} = \lambda q, \quad (1.28)$$

with solutions

$$p(t) = p_0 e^{-\lambda t}, \quad q(t) = q_0 e^{\lambda t}. \quad (1.29)$$

A square patch is exponentially squeezed in p and stretched by the same factor in q :

$$\Delta p(t) \Delta q(t) = \Delta p(0) \Delta q(0). \quad (1.30)$$

Its divergence verifies the result locally,

$$\frac{\partial \dot{p}}{\partial p} + \frac{\partial \dot{q}}{\partial q} = -\lambda + \lambda = 0. \quad (1.31)$$

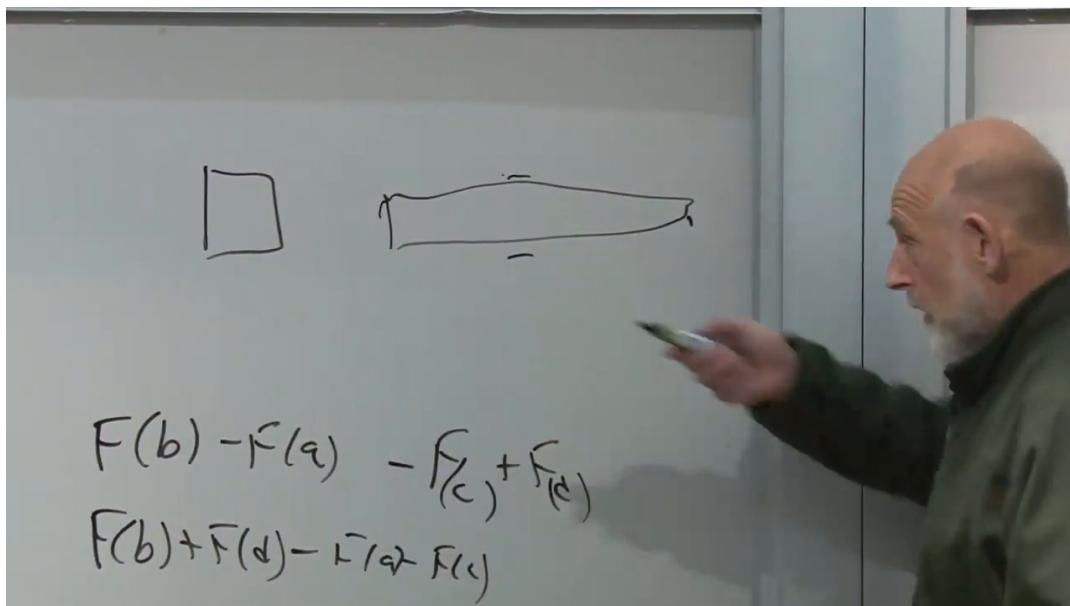


Figure 1.12: A compact patch stretched into a long thin strip while preserving area under the $H = pq$ flow.

Lecture frame: 01:02:10.

Question from the audience. Since pq has units of action, is Liouville's theorem conservation of action? ^a

Response. No. The theorem conserves phase-space volume. In one canonical pair, $dq dp$ has units of action, and that dimensional fact foreshadows quantum mechanics, where phase-space cells are measured in units set by $\hbar$. But the product pq is not itself Hamilton's action functional, nor is it automatically angular momentum merely because the dimensions agree.

^aLecture timestamp: 01:03:00–01:04:09.

The example also shows that angles need not be preserved. A right-angle cross with unequal components under the scaling in Eq. (1.29) is sheared in its metric appearance, while area remains unchanged. Liouville's theorem is an area or volume theorem, not an ordinary Euclidean rigidity theorem.

1.9 Damping Produces a Sink

To see what the theorem excludes, consider the reduced equation for a damped oscillator,

$$m\ddot{x} = -kx - c\dot{x}, \quad c > 0. \quad (1.32)$$

Let $p = m\dot{x}$. Then

$$\dot{x} = \frac{p}{m}, \quad \dot{p} = -kx - \frac{c}{m}p. \quad (1.33)$$

Without damping, the phase point circles the origin. With damping, energy is transferred into the microscopic degrees of freedom of the surrounding medium, and the orbit spirals inward.

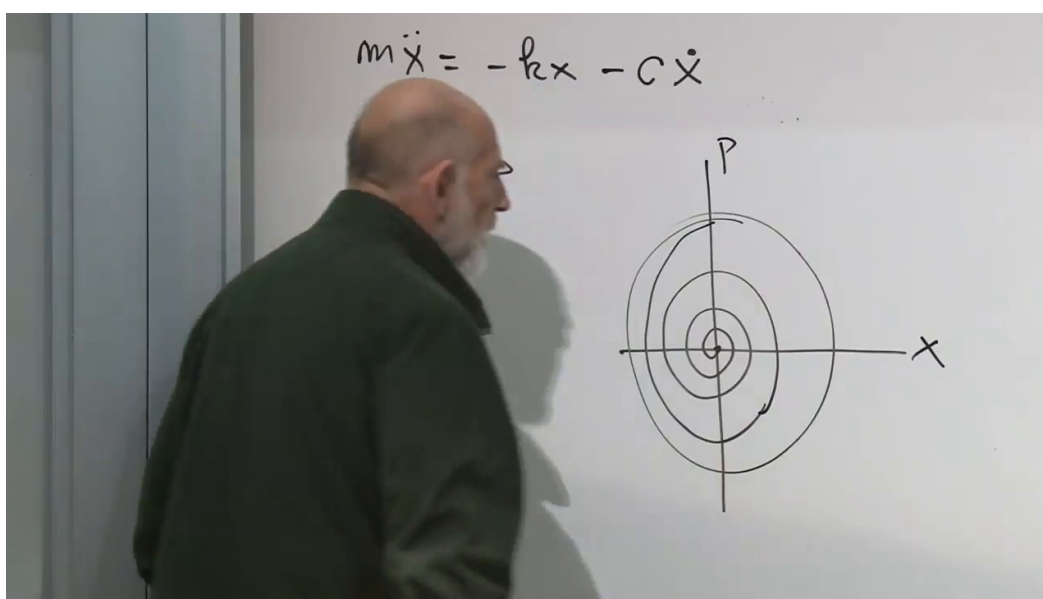


Figure 1.13: The damped oscillator equation and its inward spiral in the x - p plane.

Lecture frame: 01:13:30.

The divergence of this reduced two-dimensional flow is

$$\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{p}}{\partial p} = 0 - \frac{c}{m} < 0. \quad (1.34)$$

Every finite patch contracts. The origin is a sink, the continuous analogue of the many-arrows-in state map.

This does not mean that microscopic mechanics violates Liouville's theorem. The two variables x, p omit the enormous number of environmental degrees of freedom that receive the oscillator's energy. The enlarged oscillator-plus- environment system can be Hamiltonian and volume preserving even though its reduced x, p description contracts.

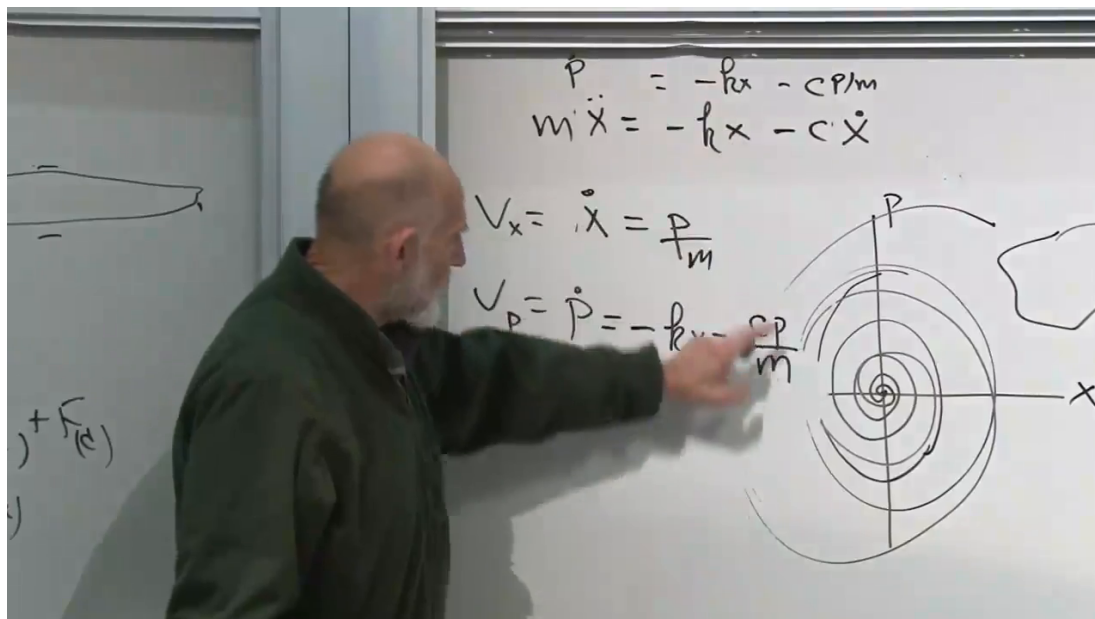


Figure 1.14: The damped phase-space velocity components used to obtain the negative divergence $-c/m$.

Lecture frame: 01:16:05.

Question from the audience. Is the damped oscillator still described by Newton's equation, even though it is not Hamiltonian in this reduced phase space? ^a

Response. Yes. Newton's $F = ma$ can include an empirically supplied drag force. The elementary $L = T - V$ and Hamiltonian description applies directly to conservative forces, and more general Lagrangians also handle some velocity-dependent forces. A Markovian friction term for an isolated pair x, p , however, is reduced dissipative dynamics, not a closed canonical Hamiltonian system.

^aLecture timestamp: 01:20:09–01:21:12.

Question from the audience. If the damped equations can be integrated backward with infinite precision, is information really lost? ^a

Response. For any finite time the ideal differential equation still has an inverse, so exact data can be run backward. The precise statement is that finite phase-space cells contract exponentially. Recovering their past then requires exponentially increasing precision, and at infinite time all trajectories approach the sink. This is the closest continuous classical analogue of the discrete loss-of-alternatives picture, not an assertion that two exact finite-time solutions literally become one.

^aLecture timestamp: 01:25:16–01:27:17.

Question from the audience. Can a vector field have zero divergence without coming from a Hamiltonian? ^a

Response. Certainly. Hamiltonian form is sufficient for zero divergence, not necessary. An arbitrary incompressible velocity field can have zero divergence without being a canonical Hamiltonian field on that same space. A three-dimensional incompressible fluid velocity field is the classroom example: three coordinates cannot themselves form canonical pairs. Full fluid mechanics may possess more sophisticated infinite-dimensional Hamiltonian formulations, but that does not reverse the one-way implication being made here.

^aLecture timestamp: 01:18:21–01:22:34.

1.10 Further Distinctions Exposed by Questions

Question from the audience. How is motion on a constant-energy surface related to preservation of a whole phase-space volume? ^a

Response. They are independent consequences of the Hamiltonian structure. If H is time independent, each trajectory remains on its own energy surface. A finite patch may cross many such surfaces; Liouville's theorem says that the volume of that entire patch is preserved. For the harmonic oscillator the patch also keeps its shape because the flow is a rigid rotation. The $H = pq$ patch keeps only its area, not its shape.

^aLecture timestamp: 01:28:47–01:30:41.

Question from the audience. Where does mechanical work fit into the Hamiltonian account? ^a

Response. Work describes energy transferred across a chosen system boundary. An external agent can raise a weight or change a parameter and thereby change the subsystem energy; the energy it supplies is lost by the agent. If both sides are included in one closed model, the emphasis returns to total energy conservation and exchange between subsystems.

^aLecture timestamp: 01:30:44–01:32:57.

Question from the audience. Does volume preservation also preserve topology and angles? ^a

Response. A smooth finite-time Hamiltonian flow preserves topology because it is an invertible continuous deformation. It does not generally preserve Euclidean angles or shape. The oscillator's rigid rotation does, but the $H = pq$ stretching demonstrates why that is a special case.

^aLecture timestamp: 01:33:00–01:33:37.

Question from the audience. Was the Legendre transformation omitted because it has no physical significance? ^a

Response. No. It is the useful transformation that exchanges the velocity description for the momentum description and produces $H = \sum_i p_i \dot{q}_i - L$ when the map is regular. The construction has already been used repeatedly even when its formal name was not emphasized; the detailed general theory is simply not needed for the present theorem.

^aLecture timestamp: 01:33:43–01:34:48.

1.11 Poisson Brackets: The Next Compression

The final minutes return to the historical theme. Nineteenth-century mechanics repeatedly searched for compact structures linking apparently different systems—mechanics and geometrical optics, particles and rays, classical trajectories and wave-like descriptions. Poisson brackets are another such structure, and they become especially important in quantum mechanics.

Let $F(q, p)$ be any function on phase space. It might be a coordinate, a momentum, the product qp , or angular momentum. Along the motion,

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i \right). \quad (1.35)$$

Substitute Hamilton's equations:

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial H}{\partial q_i} \right). \quad (1.36)$$

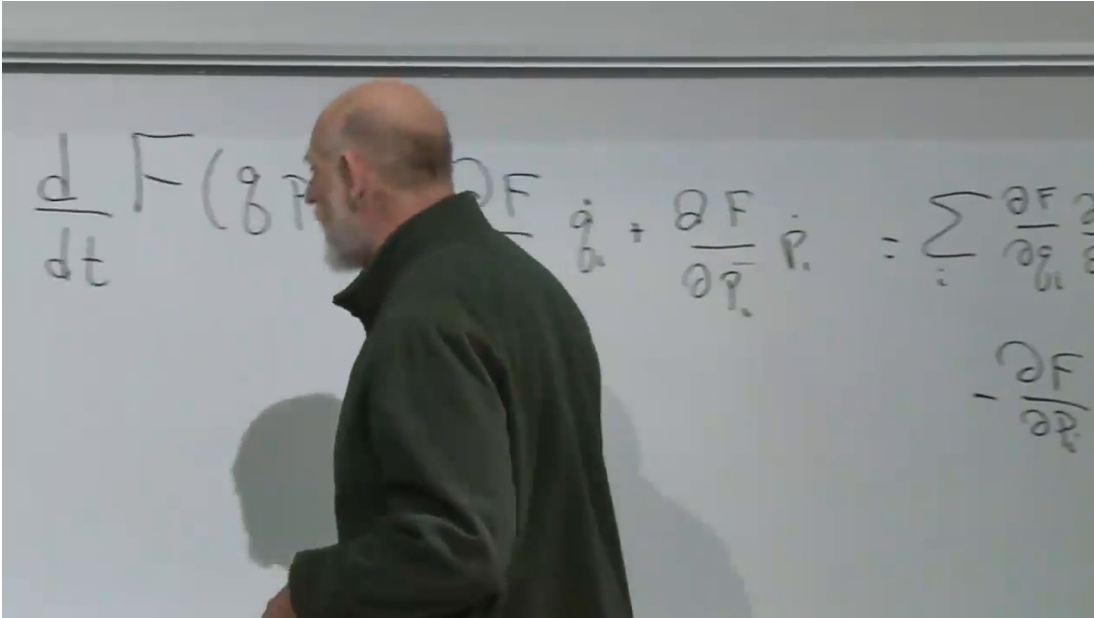


Figure 1.15: The chain rule for the time derivative of a general phase-space function $F(q, p)$.

Lecture frame: 01:42:40.

The repeated antisymmetric combination deserves a name. For any two functions F and G , define their Poisson bracket by

$$\{F, G\} = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right). \quad (1.37)$$

Equation (1.36) becomes

$$\frac{dF}{dt} = \{F, H\}. \quad (1.38)$$

If F also depends explicitly on time, one adds $\partial F / \partial t$. Taking $F = q_i$ or $F = p_i$ recovers Hamilton's two equations, so Eq. (1.38) is a compact form of the complete dynamics.

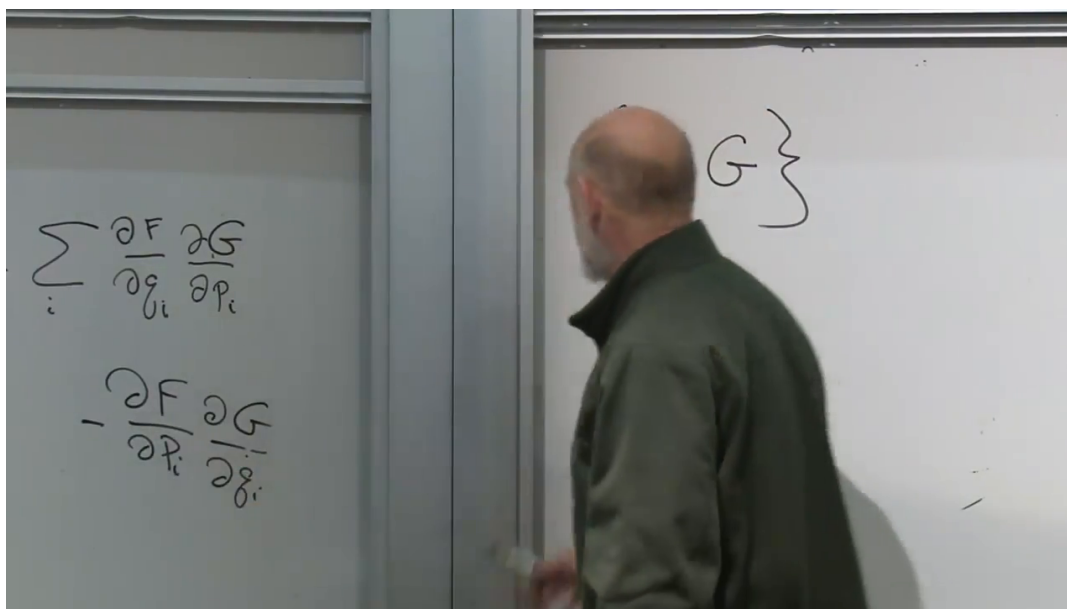


Figure 1.16: The two-function derivative combination as it is promoted to the Poisson bracket.

Lecture frame: 01:44:15.

Notation can always be compressed artificially—one could square many equations, add them, and call the result one equation. That would hide rather than reveal structure. The Poisson bracket is useful for the opposite reason: its algebra organizes time evolution, conserved quantities, and symmetry. The next step is to develop that algebra and then use it for angular momentum, whose structure will be central both in classical mechanics and in quantum theory.

1.12 The Continuous Meaning of Reversibility

We can now state the chain of ideas without shortening it. Hamilton's equations conserve H when the Hamiltonian has no explicit time dependence, so individual trajectories remain on constant-energy hypersurfaces. The same equations assign a velocity to every point of phase space. The divergence of that velocity vanishes because mixed partial derivatives cancel pairwise. Consequently, a whole region of possible states preserves its canonical volume.

That is the continuous version of one arrow in and one arrow out. Shape can change, angles can change, and neighboring trajectories can separate enormously. What cannot happen in a smooth closed Hamiltonian description is the compression of a finite collection of alternatives into a smaller phase-space volume. The $H = pq$ flow demonstrates the freedom to stretch; the reduced damped oscillator demonstrates the sink that appears once the Hamiltonian conditions are abandoned. Poisson brackets then condense the same dynamics into the statement that the Hamiltonian generates every time derivative.